

Results Tables (fill in with YOUR results)

ResNet-50 Metric Result

Training Accuracy	1.000
Validation Accuracy	0.967
Test Accuracy	1.000
Precision (macro)	1.000
Recall (macro)	1.000
F1-score (macro)	1.000

U-Net Metric Result

Training Loss	0.1451
Validation Loss	0.1944
Test Dice	0.9324
Test IoU	0.8789

Model	Dataset	Task	Main Metric	Result
ResNet-50	Oxford Flowers-102 (5 classes, 200 images)	Classification	Accuracy	1.000
U-Net	Oxford-IIIIT Pet (150 image/mask pairs)	Segmentation	Dice / IoU	0.9324 / 0.8789

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# (7a) freeze the backbone, keep only the new classifier trainable
for p in resnet.parameters():
    p.requires_grad = False
for p in resnet.fc.parameters():
    p.requires_grad = True

total, trainable = count_params(resnet)
print(f"Phase 1 -> trainable parameters: {trainable:,} of {total:,}")

# (7b) optimizer
optimizer = optim.AdamW(filter(lambda p: p.requires_grad, resnet.parameters()),
                          lr=1e-3, weight_decay=1e-4)

# Training (given)
phase1_best = fit_cls(resnet, optimizer, epochs=10, ckpt_path=os.path.join(CKPT_DIR,
print(f"Best validation accuracy after Phase 1: {phase1_best:.3f}")
```

```
... Phase 1 -> trainable parameters: 10,245 of 23,518,277
Epoch 01/10 | train loss 1.1669 acc 0.793 | val loss 1.0572 acc 0.867 | 2.1s
Epoch 02/10 | train loss 0.9158 acc 0.914 | val loss 0.8868 acc 0.900 | 2.2s
Epoch 03/10 | train loss 0.7535 acc 0.936 | val loss 0.7584 acc 0.867 | 3.0s
Epoch 04/10 | train loss 0.6424 acc 0.943 | val loss 0.6687 acc 0.867 | 2.0s
Epoch 05/10 | train loss 0.5518 acc 0.936 | val loss 0.5957 acc 0.900 | 2.0s
Epoch 06/10 | train loss 0.4861 acc 0.950 | val loss 0.5408 acc 0.900 | 2.1s
Epoch 07/10 | train loss 0.4272 acc 0.943 | val loss 0.5012 acc 0.900 | 2.1s
Epoch 08/10 | train loss 0.3817 acc 0.964 | val loss 0.4648 acc 0.900 | 2.1s
Epoch 09/10 | train loss 0.3609 acc 0.943 | val loss 0.4361 acc 0.900 | 3.0s
Epoch 10/10 | train loss 0.3121 acc 0.971 | val loss 0.4124 acc 0.900 | 2.8s
Best validation accuracy after Phase 1: 0.900
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# (8a) unfreeze layer4
for p in resnet.layer4.parameters():
    p.requires_grad = True

total, trainable = count_params(resnet)
print(f"Phase 2 -> trainable parameters: {trainable:,} of {total:,}")

# (8b) new optimizer with a smaller learning rate
optimizer = optim.AdamW(filter(lambda p: p.requires_grad, resnet.parameters()),
                          lr=1e-4, weight_decay=1e-4)

# Scheduler + training (given)
scheduler = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=10)
phase2_best = fit_cls(resnet, optimizer, epochs=10, ckpt_path=os.path.join(CKPT_DIR, "resnet_phase2_best.pth"))
print(f"Best validation accuracy after Phase 2: {phase2_best:.3f}")

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... Phase 2 -> trainable parameters: 14,974,981 of 23,518,277
Epoch 01/10 | train loss 0.7714 acc 0.707 | val loss 0.3669 acc 0.967 | 4.5s
Epoch 02/10 | train loss 0.4187 acc 0.957 | val loss 0.3120 acc 0.967 | 3.7s
Epoch 03/10 | train loss 0.2057 acc 1.000 | val loss 0.2243 acc 0.967 | 2.2s
Epoch 04/10 | train loss 0.1006 acc 0.993 | val loss 0.1852 acc 0.967 | 2.1s
Epoch 05/10 | train loss 0.0553 acc 1.000 | val loss 0.1531 acc 0.967 | 2.2s
Epoch 06/10 | train loss 0.0360 acc 1.000 | val loss 0.1475 acc 0.967 | 2.1s
Epoch 07/10 | train loss 0.0387 acc 1.000 | val loss 0.1485 acc 0.967 | 3.0s
Epoch 08/10 | train loss 0.0235 acc 1.000 | val loss 0.1523 acc 0.967 | 2.9s
Epoch 09/10 | train loss 0.0492 acc 0.993 | val loss 0.1499 acc 0.967 | 2.1s
Epoch 10/10 | train loss 0.0301 acc 1.000 | val loss 0.1495 acc 0.967 | 2.2s
Best validation accuracy after Phase 2: 0.967

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# (17a) freeze the encoder
for p in unet.encoder.parameters():
    p.requires_grad = False

total, trainable = count_params(unet)
print(f"Phase 1 -> trainable parameters: {trainable:,} of {total:,}")

# (17b) optimizer
optimizer = optim.AdamW(filter(lambda p: p.requires_grad, unet.parameters()),
                          lr=1e-3, weight_decay=1e-4)

UNET_PHASE1_BEST = fit_seg(unet, optimizer, epochs=15, ckpt_path=os.path.join(CKPT_DIR, "UNET_PHASE1_BEST.pth"))
print(f"Best validation Dice after Phase 1: {UNET_PHASE1_BEST:.3f}")

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Phase 1 -> trainable parameters: 9,013,073 of 32,521,105
Epoch 01/15 | train loss 0.8938 dice 0.772 | val loss 1.1568 dice 0.795 iou 0.669 | 32.4s
Epoch 02/15 | train loss 0.6094 dice 0.857 | val loss 0.4450 dice 0.899 iou 0.821 | 3.1s
Epoch 03/15 | train loss 0.4806 dice 0.884 | val loss 0.3523 dice 0.916 iou 0.849 | 3.1s
Epoch 04/15 | train loss 0.4017 dice 0.900 | val loss 0.3238 dice 0.922 iou 0.858 | 4.1s
Epoch 05/15 | train loss 0.3782 dice 0.894 | val loss 0.4528 dice 0.885 iou 0.799 | 2.9s
Epoch 06/15 | train loss 0.3451 dice 0.897 | val loss 0.3488 dice 0.906 iou 0.833 | 2.6s
Epoch 07/15 | train loss 0.2950 dice 0.914 | val loss 0.2469 dice 0.933 iou 0.877 | 3.1s
Epoch 08/15 | train loss 0.2805 dice 0.914 | val loss 0.2634 dice 0.920 iou 0.855 | 3.0s
Epoch 09/15 | train loss 0.2722 dice 0.915 | val loss 0.2577 dice 0.929 iou 0.871 | 3.7s
Epoch 10/15 | train loss 0.2557 dice 0.923 | val loss 0.2626 dice 0.927 iou 0.866 | 3.0s
Epoch 11/15 | train loss 0.2677 dice 0.917 | val loss 0.2683 dice 0.925 iou 0.864 | 3.4s
Epoch 12/15 | train loss 0.2575 dice 0.919 | val loss 0.2539 dice 0.926 iou 0.867 | 2.7s
Epoch 13/15 | train loss 0.2538 dice 0.919 | val loss 0.2503 dice 0.929 iou 0.870 | 3.2s
Epoch 14/15 | train loss 0.2397 dice 0.925 | val loss 0.2435 dice 0.930 iou 0.872 | 3.2s
Epoch 15/15 | train loss 0.2104 dice 0.932 | val loss 0.2212 dice 0.935 iou 0.881 | 3.0s
Best validation Dice after Phase 1: 0.935

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for p in unet.encoder.layer4.parameters():
    p.requires_grad = True

total, trainable = count_params(unet)
print(f"Phase 2 -> trainable parameters: {trainable:,} of {total:,}")

# (18b) optimizer with two parameter groups
enc_params = list(unet.encoder.layer3.parameters()) + list(unet.encoder.layer4.parameters())
enc_ids = {id(p) for p in enc_params}
other_params = [p for p in unet.parameters() if p.requires_grad and id(p) not in enc_ids]

optimizer = optim.AdamW([
    {"params": enc_params, "lr": 1e-4}, # unfrozen encoder layers
    {"params": other_params, "lr": 3e-4}, # decoder + segmentation head
], weight_decay=1e-4)

scheduler = optim.lr_scheduler.CosineAnnealingLR(optimizer, T_max=15)
UNET_PHASE2_BEST = fit_seg(unet, optimizer, epochs=15, ckpt_path=os.path.join(CKPT_DIR, "UNET_BE
print(f"Best validation Dice after Phase 2: {UNET_PHASE2_BEST:.3f}")

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Phase 2 -> trainable parameters: 31,076,177 of 32,521,105
Epoch 01/15 | train loss 0.2755 dice 0.913 | val loss 0.2303 dice 0.935 iou 0.880 | 6.1s
Epoch 02/15 | train loss 0.2275 dice 0.927 | val loss 0.2255 dice 0.933 iou 0.877 | 3.5s
Epoch 03/15 | train loss 0.1924 dice 0.936 | val loss 0.2121 dice 0.938 iou 0.885 | 3.6s
Epoch 04/15 | train loss 0.2112 dice 0.931 | val loss 0.2077 dice 0.940 iou 0.889 | 3.6s
Epoch 05/15 | train loss 0.1892 dice 0.937 | val loss 0.2366 dice 0.933 iou 0.877 | 3.5s
Epoch 06/15 | train loss 0.1780 dice 0.941 | val loss 0.1912 dice 0.944 iou 0.896 | 4.0s
Epoch 07/15 | train loss 0.1802 dice 0.939 | val loss 0.2016 dice 0.942 iou 0.893 | 3.3s
Epoch 08/15 | train loss 0.1588 dice 0.947 | val loss 0.2015 dice 0.943 iou 0.894 | 3.3s
Epoch 09/15 | train loss 0.1632 dice 0.946 | val loss 0.2019 dice 0.943 iou 0.894 | 3.7s
Epoch 10/15 | train loss 0.1528 dice 0.948 | val loss 0.1996 dice 0.945 iou 0.898 | 4.1s
Epoch 11/15 | train loss 0.1493 dice 0.950 | val loss 0.2038 dice 0.943 iou 0.895 | 3.4s
Epoch 12/15 | train loss 0.1475 dice 0.949 | val loss 0.2000 dice 0.943 iou 0.895 | 3.3s
Epoch 13/15 | train loss 0.1431 dice 0.951 | val loss 0.2004 dice 0.943 iou 0.895 | 3.6s
Epoch 14/15 | train loss 0.1575 dice 0.952 | val loss 0.1937 dice 0.945 iou 0.898 | 7.3s
Epoch 15/15 | train loss 0.1451 dice 0.951 | val loss 0.1944 dice 0.945 iou 0.898 | 3.2s
Best validation Dice after Phase 2: 0.945

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